

ASSESSMENT TOOL
SHORT ANSWER TEST

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DEPARTMENT: CSE(AI)

28. A digital image has reduced contrast due to poor sensing conditions. Write a Python program to enhance a low-contrast image.

PROGRAM:-

```
import cv2

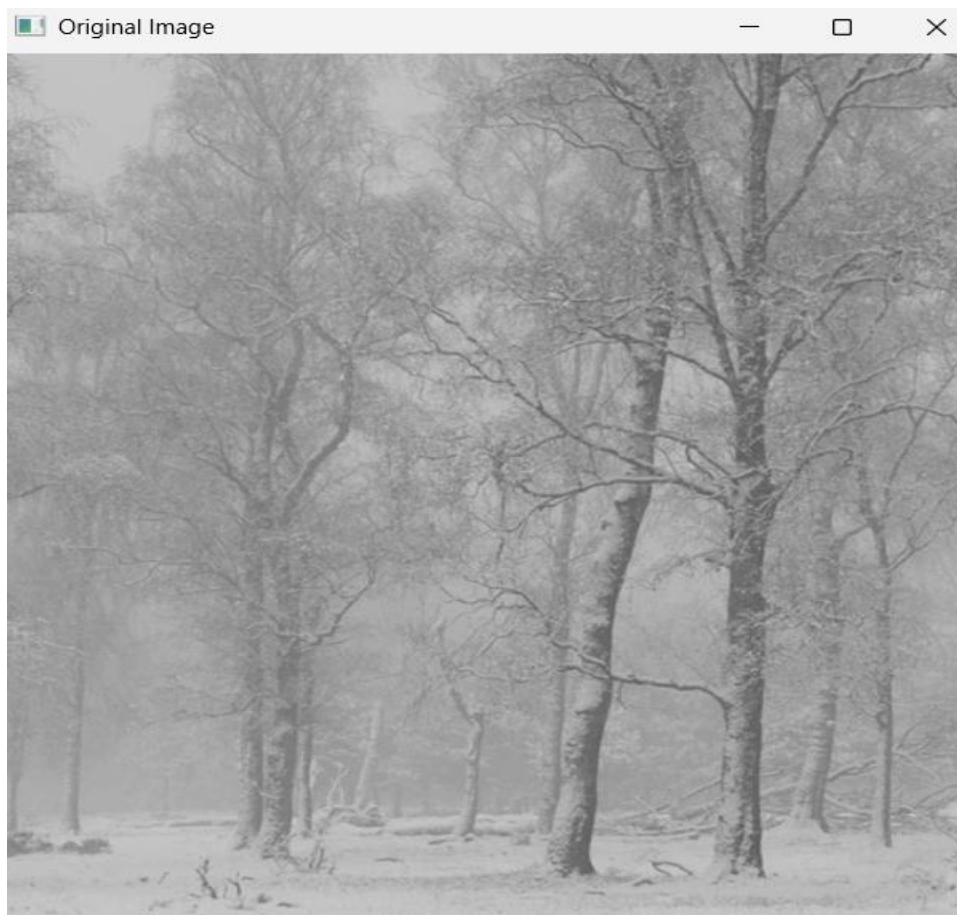
img = cv2.imread(r"C:\Users\Hema Neepe\OneDrive\Pictures\Screenshots\Screenshot 2026-07-24 094307.png", 0)

enhanced = cv2.equalizeHist(img)

cv2.imshow("Original Image", img)
cv2.imshow("Enhanced Image", enhanced)

cv2.waitKey(0)
cv2.destroyAllWindows()
cv2.imwrite("enhanced_image.jpg", enhanced)
```

OUTPUT:-





29. A machine vision system detects product defects. Write a Python program for defect detection.

PROGRAM:-

```
import cv2

img = cv2.imread(r"C:\Users\Hema Neepa\OneDrive\Pictures\Screenshots\Screenshot 2026-07-24 095359.png")

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

_, th = cv2.threshold(gray, 120, 255, cv2.THRESH_BINARY_INV)

contours, _ = cv2.findContours(th, cv2.RETR_EXTERNAL,

cv2.CHAIN_APPROX_SIMPLE)
```

for c in contours:

if cv2.contourArea(c) > 50:

x, y, w, h = cv2.boundingRect(c)

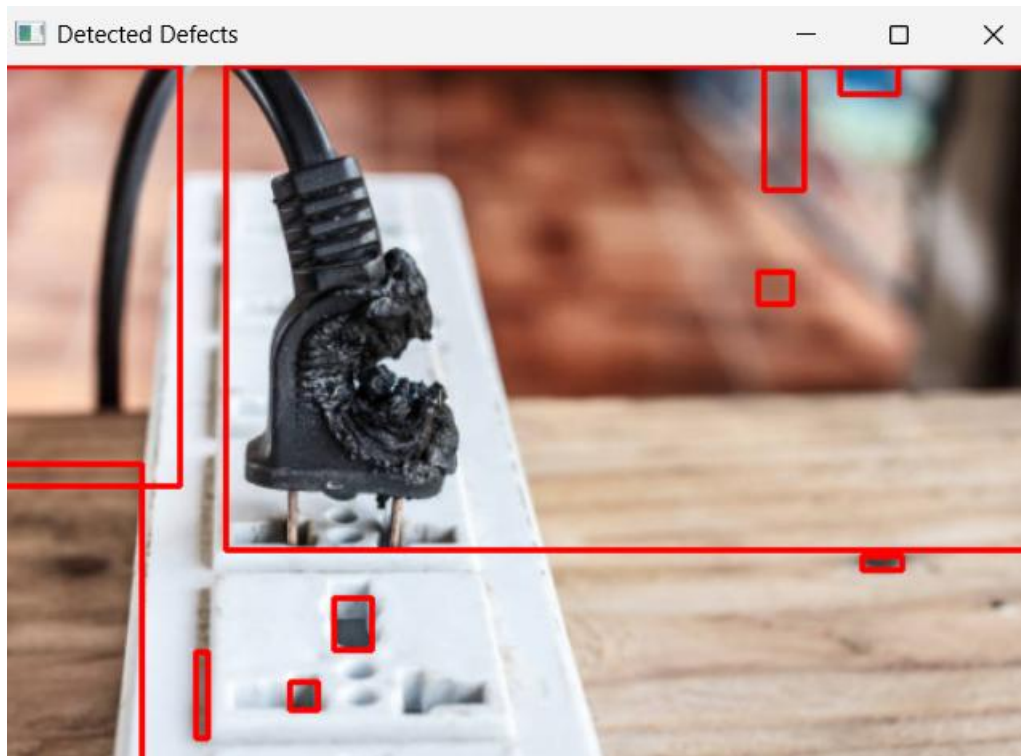
cv2.rectangle(img, (x, y), (x + w, y + h), (0, 0, 255), 2)

cv2.imshow("Detected Defects", img)

cv2.waitKey(0)

cv2.destroyAllWindows()

OUTPUT:-



30. A camera captures images with varying brightness levels. Write a Python program to adjust image brightness.

PROGRAM:-

```
import cv2

img = cv2.imread(r"C:\Users\Hema Neeva\OneDrive\Pictures\Screenshots\Screenshot 2026-07-24 101924.png")

bright = cv2.convertScaleAbs(img, alpha=1, beta=50)

cv2.imshow("Original Image", img)

cv2.imshow("Brightened Image", bright)

cv2.waitKey(0)

cv2.destroyAllWindows()

cv2.imwrite("bright_image.jpg", bright)
```

OUTPUT:-

